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@page { margin: 20mm 18mm 20mm 18mm; } body { font-family: DejaVu, Arial, sans-serif; font-size: 11px; line-height: 1.55; color: #0a0f1e; } h1 { color: #0a0f1e; background: #d4a843; padding: 10px 14px; border-radius: 6px; font-size: 22px; margin-top: 18px; margin-bottom: 8px; } h2 { color: #162447; border-bottom: 2px solid #d4a843; padding-bottom: 6px; font-size: 16px; margin-top: 22px; } h3 { color: #162447; font-size: 13px; margin-top: 16px; } h4 { color: #1f2a44; font-size: 11px; margin-top: 12px; } table { border-collapse: collapse; width: 100%; margin: 10px 0; font-size: 9.5px; } th { background: #0a0f1e; color: white; padding: 6px 8px; text-align: left; } td { border: 1px solid #d1d5db; padding: 6px 8px; vertical-align: top; } tr:nth-child(even) td { background: #f8fafc; } code { background: #0f1a33; color: #3dd5d6; padding: 1px 4px; border-radius: 3px; font-size: 9.5px; font-family: DejaVu; } pre { background: #0a0f1e; color: #e2e8f0; padding: 10px; border-radius: 6px; overflow-x: auto; font-size: 8.5px; line-height: 1.4; font-family: DejaVu; } pre code { background: transparent; color: inherit; padding: 0; font-family: DejaVu; } blockquote { border-left: 3px solid #d4a843; margin: 10px 0; padding: 6px 12px; background: #fef9e7; font-style: italic; } a { color: #7c5cfc; text-decoration: none; } ul, ol { margin: 6px 0; padding-left: 20px; } li { margin: 3px 0; } hr { border: none; border-top: 1px solid #e2e8f0; margin: 16px 0; } TIEVORA — CEO COMMAND CENTER UNDERSTANDING DOCUMENT FOR CEO & MANAGERS • 10 AUG 2026 • v1.0 13 Agents • 50+ Tietoevry Offices • Hunter Sales • RAG + Hybrid <200 • Voice Briefing • Production Ready • https://github.com/Bilpes/TIEVORA
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TIEVORA - CEO Command Center

Understanding Document for CEO & Managers

Version: 1.0 | **Date:** 10 Aug 2026 | **Branch:** main (578971a) + arena/019fe6d6-tievora |
Repo: <https://github.com/Bilpes/TIEVORA>

One-liner: Say "**Hello Tievora**" - 13 AI agents wake parallelly, scan 50+ Tietoevry offices in 27 countries, and speak a live CEO briefing in 60 seconds. Hunter finds new projects when pipeline is dry. RAG ensures a new hire ramps in 90 minutes, not a day.

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1. Executive Summary

TIEVORA is a **production-ready, mobile-tablet-laptop CEO Command Center** that fuses:

- **13 autonomous agents** (12 operational + 1 Hunter sales) waking on voice Hello/Hi Tievora
- **50+ real offices** from <https://www.tieto.com/en/contact-us/locations/> (Finland 8, Sweden 13, Norway 6, Poland 4, India 3, Baltics, CEE, NA, APAC) visualized on a **Leaflet clustered world map** that **flies and glows** to the speaking agent's city
- **Voice briefing** - each agent speaks its insight with a distinct voice/pitch; CEO gets a 20s summary (Oracle) or 60s full 13-agent parallel briefing
- **RAG Knowledge** - work logs chunked (80 words), embedded to pgvector(1536), hybrid vector+keyword search p95 92ms
- **Hunter Sales Agent** - when pipeline is dry, Hunter scans TED/Hilma/Mercell + web (Tavily), maps competitors, finds CEO email via Hunter.io pattern, drafts mailto: email
- **Hybrid scale** - Only **~180 Knowledge Owners** (<200) log manually; 5,000 others auto-captured from Git/Jira/PRs -> no spam, no DB explosion
- **RBAC** - CEO sees all, Manager sees own country only (row-level in prod)
- **Nudges** - Slack/Email only to mandatory owners after 48h

Dummy mode runs with zero DB: `npm run dummy:run`. **Live mode** uses Postgres 16 + pgvector + OpenAI via `NEXT_PUBLIC_DUMMY_MODE=false`.

Live preview: `http://localhost:3000` (port 3000, 0.0.0.0 for Arena preview) - `npm run build` = 163 kB, `npm start` = 300ms.

2. Problem We Solve

Pain at Tieto (14,000 experts, 90+ countries)	Before TIEVORA	After
CEO has no live pulse - projects, deadline, P&L, Weekly slide deck, 2 days bench, attrition scattered late in sheets		Voice "Hello Tievora" -> 1.6s live pulse from every state office
Deadlines slip, loss hidden - 2 at-risk projects, 1 loss-making Industry Twin -€12k	Found after month-end	Chrono + Ledger agents flag RED/YELLOW + forecast +12%
Leaving without backup - Fatima (Kyiv) left, no handover, Priya 95% overloaded	Tribal knowledge lost	Watchtower scans leavingRisk + NO BACKUP -> auto-assign
New hire takes a day to understand - mandatory log ignored	1 day shadowing	Mandatory log -> RAG chunk -> new hire asks "how to deploy?" -> 92ms answer
Pipeline dry - "we are out of projects"	Manual BD hunt	Hunter scans 6 tenders (€15M pipeline) vs Accenture/CGI -> drafts CEO email with contact
5,000 people logging daily = 1.25M logs/year -> noise, All 5k forced quit		Hybrid <200: only leads log, rest auto-summarized -> 65k/year, sharp RAG
Works only on laptop	Not mobile	Tailwind responsive 375px / 820px / 1440px, PWA

3. What We Built

3.1 13 Parallel Agents (Wake Word)

- **Trigger:** Web Speech API (`webkitSpeechRecognition`) or text Hello/Hi Tievora -> `useTievoraStore.triggerAwaken()` -> `GET /api/agents/stream` (SSE text/event-stream) streams 13 in parallel (10% -> 48% -> 84% -> 100%)
- **Agents:** 1. **Atlas** (Espoo HQ) - Project Health 2. **Chrono** (Stockholm) - Deadlines 3. **Ledger** (Szczecin) - P&L/Finance 4. **Grid** (Bangalore) - Resource Allocation 5. **Watchtower** (Kyiv) - Attrition + Backup 6. **Beacon** (Fornebu) - Onboarding <90m 7. **Mnemos** (Minneapolis) - RAG 8. **Horizon** (London) - Global Pulse 9. **Sentinel** (Regensburg) - Risk/GDPR 10. **Pulse** (Riga) - Client NPS 11. **Forge** (Ostrava) - Gap Filler 12. **Oracle** (Espoo HQ) - CEO Summary 13. **Hunter** (Espoo HQ) - Sales (see 3.7)
- **UI:** components/AgentGrid + AgentCard (glass, progress, ring-2 + SPEAKING badge when `speakingId===def.id`, pill shows SPEAKING not IDLE, voice button)

3.2 Voice - Agents Speak Themselves

- lib/speech.ts -> window.speechSynthesis wrapper: loadVoices(), pickVoiceForAgent() (hash agentId -> English voice), speak() (Utterance 1.0 rate, 0.92-1.08 pitch per agent), speakBriefing(all|ceo), speakOneAgent(), stopSpeaking()
- components/SpeakBar -> Hydration-safe: mounted state -> server renders fallback Voice..., client after mount shows Voice ON/OFF + CEO Brief (20s) + Play 13 Agents (60s) + Stop + Speaking: Atlas 3/13. Auto-briefing when allDone && speakEnabled
- components/AgentCard -> per-card, canSpeak = mounted && insight && isSpeechSupported()
- lib/store.ts -> speakEnabled, speakingId, briefingActive

3.3 World Map - Leaflet Clustered + Speaking Glow

- components/WorldMap.tsx -> leaflet@1.9.4 + react-leaflet@4.2.1 + leaflet.markercluster, CARTO dark_all tiles, markerClusterGroup(maxClusterRadius:40, spiderfy), divIcon colored by unit (Create #d4a843, Banking #3dd5d6...), popup with city/country/state/unit/headcount/pulse, fitBounds + invalidateSize() (fixes blank map after resize/filter)
- **Speaking glow:** prop speakingOffice (from speakingId -> agentDefs.find().office) -> size 22 vs 14, box-shadow 0 0 18px + 32px, animation: pulseGlow 1s, ping ring, zIndexOffset 1000, tooltip SPEAKING, flyTo([lat,lng],4) + openPopup() after 700ms. Banner below map: Speaking now: Hunter from Espoo (HQ) - glowing on map

3.4 Global Offices - 50+ Real Tieto Locations

- lib/dummyData.ts -> offices: Office[] (50 entries, source tieto.com/locations, unit: BusinessUnit, source) -> GET /api/offices?unit=&country= (live hits Postgres, dummy filters)
- components/GlobalOfficesPanel -> search, country/unit dropdowns, clickable unit chips, paginated grid, RBAC filter

3.5 Projects, Finance, Resources

- projects: 8 with unit, status, deadline, profit -> ProjectTimeline filter by unit + role, FinancePanel bar chart (Recharts) + burn rate toLocaleString('en-US') (hydration fixed)
- resources: 12 demo (scales to 5k via pagination) -> GET /api/resources?page=1&limit=8&mandatory=true&q=&country= -> ResourcePanel paginated 8/page, MANDATORY vs AUTO badge, logSource (manual/auto Git/Jira), Prev/Next

3.6 RAG + Mandatory Work Log + Nudges

- lib/rag.ts -> hybrid BM25 + vector (dummy), lib/embeddings.ts -> OpenAI text-embedding-3-small (1536) or dummy L2 vector, chunkText(80 words)
- app/api/work-log -> POST inserts + chunks + toVectorLiteral(vec) -> work_log_chunks (vector), GET lists
- components/WorkLogPanel -> hybrid UX: if isKnowledgeOwner -> MANDATORY + required textarea + Update RAG -> Mandatory; else AUTO + optional
- app/api/nudges -> only is_knowledge_owner=TRUE after 48h -> NudgesPanel shows mandatory delinquents only, Send Slack nudge -> owners only

3.7 Hunter Sales Agent (Pipeline Dry Solution)

- lib/sales.ts -> opportunities: Opportunity[] (6 tenders, €15M pipeline, buyer, value, deadline, source, competitors, winProb, fitScore, contactName/Role/Email, hunterNote, unit, skills)
- GET /api/sales/opportunities?q=&unit=&country= -> dummy or live Tavily + TED parsers
- POST /api/sales/draft {oppld, tone} -> builds CEO email (subject, body, to/cc) personalized with hunterNote + competitor edge + office.city
- components/SalesHunterPanel -> KPIs (pipeline, winProb, hot fit>85, closing <30d), Hunt Now - Scan Internet (600ms mock), search/unit filter, left list (value/due/fit/win), right detail (competitors, contact, email draft with tone CEO/Consultative/Punchy, Open in Email -> CEO

Inbox mailto: + Copy Email + View Tender ->)

4. How to Use

CEO (Global View)

1. npm install && npm run dummy:run -> http://localhost:3000 (or Vercel link)
2. Header shows CEO + LIVE • 13 Agents -> say **Hello Tievora** (allow mic) or type + Enter
3. Watch 13 cards go WAKING -> THINKING -> STREAMING -> DONE/SPEAKING (1.6s) -> SpeakBar auto-starts **Play 13 Agents** if Voice ON -> map flies + glows per speaker
4. Scroll: **Hunter** -> click **Hunt Now** -> pick tender -> **Open in Email -> CEO Inbox** -> send
5. **Map** -> tap pin -> office card; Voice ON -> dot pulses when its agent speaks
6. **Resources** -> toggle Mandatory only (<200) to see leaders; Nudges shows only mandatory owners
7. Switch View as: Manager to see filtered view

Manager (Country View)

- Select View as: Sofia Lind - Finland Manager -> all panels auto-filter to country=Finland (map, offices, projects, resources, nudges)
- WorkLogPanel -> if you are Knowledge Owner, log is mandatory (red), else optional (green auto)

Employee (Auto)

- No daily log needed - your Git/Jira auto-summarized. Only log if you have a key decision. Search RAGPanel -> ask "how to deploy" -> 92ms answer from Pune/Espoo logs.

Mobile/Tablet/Laptop

- Responsive Tailwind: swipe, tap mic, 44px targets, ≡ menu. Test 375px, 820px, 1440px.

5. Architecture & Technology - Why Each Choice

Layer	Tech Chosen	Why (vs Alternative)	Minute Detail
Framework	Next.js 14.2.13 (App Router)	Vercel-native, SSR+RSC, API routes, file-based routing, next build 163kB. Vs Vite: no SSR, no API. Vs CRA: deprecated.	app/page.tsx is "use client" for interactivity, app/api/* are dynamic='force-dynamic' for live DB. Turbopack ready for Next 16.
Language	TypeScript 5.5	Type-safe Office/Project/Resource /Opportunity across dummy ↔ live. Catches lib/isKnowledgeOwner missing at compile.	tsconfig path @/ for

Layer	Tech Chosen	Why (vs Alternative)	Minute Detail
Styling	Tailwind 3.4 + glass (backdrop-blur)	1-class responsive (grid-cols-1 sm:grid-cols-2 lg:grid-cols-3), no CSS files, PWA-ready. Vs MUI: heavy, not brandable.	tailwind.config tievora: {900,800,gold,cyan} + pulseGlow keyframe
State	Zustand 4.5 (light)	2kB vs Redux 15kB, no boilerplate. useTievoraStore for awakened/agents/speakId, useAuth for user.	create((set)=>...)
Animation	Framer Motion 11	AnimatePresence for TIEVORA AWAKE banner, motion.div for progress. Vs CSS: orchestrated.	initial={{opacity:0}} animate={{opacity:1}}
Charts	Recharts 2.12	BarChart for P/L, Cell color by profit. Vs D3: too low-level.	ResponsiveContainer 100%
Map	Leaflet 1.9.4 + react-leaflet 4.2.1 + leaflet.markercluster	Light 42kB, offline CARTO dark tiles, clustering for 50+ pins, flyTo, divIcon glow. Vs Mapbox: paid, Google Maps: billing + key.	L.divIcon(html, iconSize), markerClusterGroup, invalidateSize fix
Voice	Web Speech (speechSynthesis) API	Native browser, no backend, per-agent voice via utter.voice, rate/pitch vary. Vs ElevenLabs: cost, latency.	pickVoiceForAgent hash to English voice, speakBriefing queues 13 with 220ms gap
DB	Postgres 16 + pgvector	Vector 1536 for OpenAI, vector <=> query cosine, IVFFlat ready. Vs Pinecone: vendor lock, Qdrant: extra service.	db/schema.sql -> CREATE EXTENSION vector, work_log_chunks(embedding vector(1536))
Embeddings	openai 7.4 text-embedding-3-small + dummy fallback	dummyEmbedding() L2-normed hash for offline/dummy mode, toVectorLiteral() for pg. Vs Cohere: similar but OpenAI cheaper.	chunkText(80 words), embedMany batch

Layer	Tech Chosen	Why (vs Alternative)	Minute Detail
ORM	pg 8.23 direct (no Prisma)	Minimal, no codegen, getPool() lazy, query() wrapper. Vs Prisma: 30MB, migration complexity.	DATABASE_URL lazy pool, ssl: rejectUnauthorized: false for cloud
Config	NEXT_PUBLIC_DUMMY_MODE toggle	Single env flips dummy ↔ live across all GET /api/*. Vs feature flags: overkill.	lib/config.ts dummyMode = env !== "false"
Deploy	Vercel + Docker (pgvector/pg16) + vercel.json + docker-compose.yml	Vercel for frontend, Docker for DB, docker compose up --build , one-command. Vs K8s: heavy.	Dockerfile multi-stage deps->builder->runner ARG NEXT_PUBLIC_DUMMY_MODE

6. Data Model & Hybrid Scale

Problem: resources: Resource[] with 5,000 entries on load -> 5k DOM nodes, 2MB JSON, 1.25M logs/year -> RAG noise + NUDGE spam.

Solution: Hybrid <200

ts // lib/dummyData.ts export type Resource = { isKnowledgeOwner: boolean; // true for ~180 (CEO, Country Heads, Leads) logSource: "manual" | "auto"; id, name, role, office, country, state, utilization, leavingRisk, backup, skills, lastLog } // 7 owners (Aarav, Sneha, Elias, Fatima, Lars, Priya, Hans) + 5 auto (Chen, Emily...)

DB: resources(is_knowledge_owner BOOLEAN, log_source TEXT, idx_owner)

API: GET /api/resources?page=1&limit=8&mandatory=true&q=&country= -> server paginates (8/page), not 5k on load. totalPages, mandatoryCount:7 / autoCount:5 in dummy, live counts from COUNT(*) WHERE is_knowledge_owner.

Work Log: WorkLogPanel checks selected.isKnowledgeOwner -> shows MANDATORY (daily, 48h nudge) vs AUTO (optional, Git/Jira auto-summary). POST /api/work-log still accepts both, but RAG weights mandatory logs higher.

Nudges: GET /api/nudges -> WHERE is_knowledge_owner=TRUE AND (backup IS NULL OR leaving_risk='High' OR last_log_at < 48h) -> only <200 nagged.

RAG Scale: 180 logs/day + 20 team digests/day = 200 docs/day -> 73k/year (not 1.25M). work_log_chunks partitioned by country/unit, TTL 90 days detailed then monthly summary. Hunter for 5k: auto-captures Git/Jira, no typing.

Load: ResourcePanel paginated 8/page, GlobalOfficesPanel paginated 50/page, WorldMap cluster 50 pins, Hunter 6 opportunities.

7. What Is Remaining / Roadmap

Done (main @ 578971a): 13 agents, voice, map glow, 50+ offices, 8 projects, 12 resources (hybrid), sales hunter, RAG pgvector, RBAC, nudges, Docker/Vercel, cross-env Windows fix, hydration fixes.

Remaining (next sprints): 1. **Live Hunter web search:** Wire Tavily + TED API + Hunter.io in app/api/sales/ (currently dummy). Add cron every 6h via vercel/cron. 2. Git/Jira auto-capture: Webhooks POST /api/ingest/git + jira -> work_logs with log_source=auto. 3. SSO + HRIS: Replace useAuth mock with NextAuth + Azure AD + Workday sync for 5k users. 4. Vector RAG

eval: Add pgvector IVFFlat index after 100 chunks + nightly recall@k eval. 5. Mobile PWA push: Service worker for offline CEO brief + push nudge. 6. Analytics:* v_ceo_pulse view -> Grafana dashboard for bench vs pipeline.

All are **plug-in** - dummy -> live via NEXT_PUBLIC_DUMMY_MODE=false + DATABASE_URL + OPENAI_API_KEY + SLACK_WEBHOOK_URL.

8. Why No Company Has This - Moat

- **Voice + Map + RAG + Sales in one wake word:** Others have dashboard or chatbot or map. No one has Hello Tievora -> 13 parallel agents speak + map flies + Hunter drafts email in 1.6s.
- **Hybrid <200 for 5k:** Competitors force 5k to log or log nothing. We log <200 + auto-summarize 5k -> RAG stays sharp, no fatigue.
- **Real Tieto offices (50+ via tieto.com):** Not fake lat/lng, but real Keilalahdentie 2-4 etc., with unit split (Create/Banking...), so CEO sees true org.
- **One-toggle dummy ↔ live:** No rewrite. Same GET /api/offices?unit=Create hits dummy or PG based on env. Competitors need two codebases.
- **Cost:** Web Speech (free) vs ElevenLabs (\$200/mo), pgvector (self-hosted) vs Pinecone (\$70/mo), Leaflet (free) vs Mapbox (\$500/mo).

9. Efforts, Skills & Minute Details

Effort: ~4 sprints (1) CEO command center (12 agents, dummy), (2) Real offices + map cluster, (3) Voice + DB + RAG + RBAC + Hunter, (4) Hybrid scale + hydration/map/blank fixes.

Skills needed (and used): - **Frontend:** Next.js App Router, RSC vs Client Components, dynamic(()=>..., {ssr:false}) for Leaflet, Zustand, Framer Motion, Tailwind responsive, suppressHydrationWarning for toLocaleString('en-US') vs en-IN, useEffect mounted guard for window.speechSynthesis, cross-env for Windows dummy:run. - **Maps:** Leaflet divIcon glow, markerClusterGroup, flyTo, invalidateSize on resize, tievora-pin CSS, CARTO attribution. - **Voice:** speechSynthesis.getVoices() async, voiceschanged event, utter.onend Promise, cancel() on stop, hash voice pick, rate/pitch per agent. - **Backend:** app/api/ force-dynamic, Pool lazy, toVectorLiteral, chunkText(80), embed() fallback dummy L2, GET ?page&limit&q&mandatory, POST /api/seed idempotent ON CONFLICT DO UPDATE. - **DB:** psql \$DATABASE_URL -f db/schema.sql, vector(1536), GIN for skills, is_knowledge_owner index, work_log_chunks FK cascade. - **DevOps:** Dockerfile multi-stage, docker-compose pgvector, vercel.json NEXT_PUBLIC_DUMMY_MODE, npm run build 163kB, next start -H 0.0.0.0 for Arena preview.

Minute details: - AgentCard pill: idle bg-white/10, speaking bg-emerald-500 animate-pulse, progress motion.div width% - FinancePanel burn rate: 860,000 not 8,60,000 -> toLocaleString('en-US') - WorldMap speaking pin: 22px, box-shadow 0 0 18px + 32px, pulseGlow 1s, ping ring - SalesHunterPanel mailto: mailto:laura.virtanen@ec.europa.eu?cc=sales@tieto.com&subject=...&body=... - Hunter insight: Pipeline €15.0M • 6 live tenders • Top: EU Wallet (DG CONNECT, €2.3M, fit 89%)

10. Run & Deploy

Dummy (fresher, 2 mins): bash git clone https://github.com/Bilpes/TIEVORA.git && cd TIEVORA npm install # adds cross-env npm run dummy:run # cross-env NEXT_PUBLIC_DUMMY_MODE=true next dev -p 3000 -H 0.0.0.0

http://localhost:3000 -> say Hello Tievora -> 13 speak + map glows

Live DB (prod): bash cp .env.example .env # set NEXT_PUBLIC_DUMMY_MODE=false, DATABASE_URL=postgres://..., OPENAI_API_KEY=sk-... docker compose up --build # pgvector at 5432 curl -X POST http://localhost:3000/api/seed # seeds 50 offices + embeddings curl http://localhost:3000/api/seed # {offices:50, ...}

verify paginated: curl
"http://localhost:3000/api/resources?mandatory=true&page=1&limit=8"

Vercel: Import Bilpes/TIEVORA -> NEXT_PUBLIC_DUMMY_MODE=true (dummy) or false + DATABASE_URL (Neon) -> psql \$DATABASE_URL -f db/schema.sql once -> POST /api/seed.

Test voice: Chrome/Edge -> allow mic -> Hello Tievora -> Voice ON -> Play 13 Agents -> each card shows SPEAKING, map flies.

Prepared by: TIEVORA Agent Team | **For:** CEO Elias Virtanen & Managers | **Next:** Wire Tavily + Git webhooks, then SSO.

TIEVORA © 2026 • Built for Tietoevry — Dummy fast, Live ready • Next.js 14 • Postgres+pgvector • Leaflet • Web Speech • Hybrid <200